Harness

File: harness.kicad_sch

Daughterboard

File: daughter.kicad_sch

LSU, MAP, IAT, USB, AUX

File: io.kicad_sch

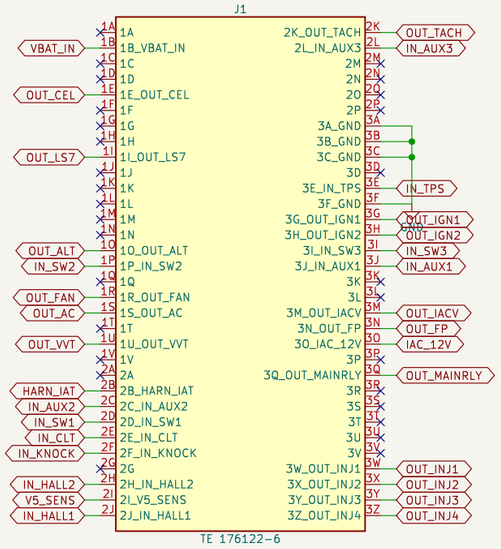
Miata ECU motherboard 1999–2000 (NB1). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

Sheet: /
File: miata_mb_9900.kicad_sch

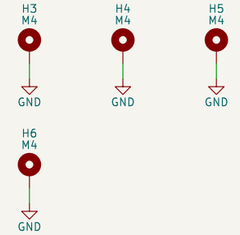
Title: Miata ECU motherboard 1999–2000 (NB1)

Size: A4	Date: 2026-10-29	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Stock harness header (Miata ECU motherboard 1999–2000): every pin labelled with the daughterboard signal it drives



Case mounting bosses (M4), tied to ground



Generated from scripts/mb_nb1.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

Title: Miata ECU motherboard 1999–2000 — Harness

Size: A3	Date: 2026-10-29	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

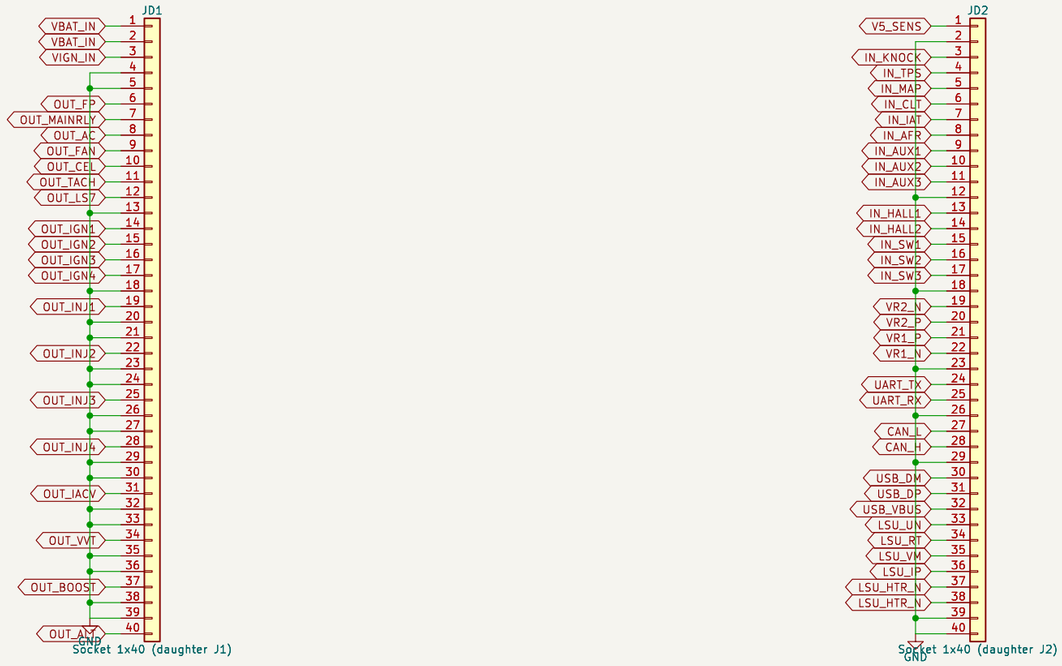
Daughterboard standoffs (M3)

HD1 M3 standoff

HD2 M3 standoff

HD3 M3 standoff

HD4 M3 standoff



Generated from scripts/mb_nb1.py; netlist identical to the routed board

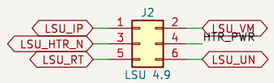
Sheet: /Daughterboard/
File: daughter.kicad_sch

Title: Miata ECU motherboard 1999–2000 — Daughterboard

Size: A3	Date: 2026-10-29	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

LSU 4.9 wideband sensor (controller on the daughterboard)

LSU 4.9: 1 IP 2 VM 3 H− 4 H+ 5 RT 6 UN

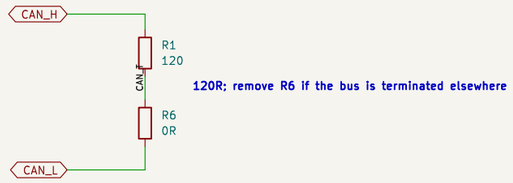




Heater fuse: 2 A hold, 30 V PTC

IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

CAN bus termination at the ECU end

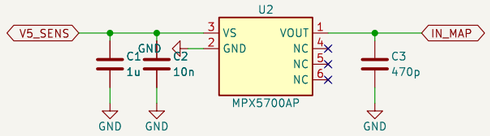


120R; remove R6 if the bus is terminated elsewhere

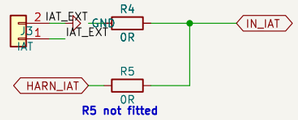
Fused +12 V feeds to harness actuators



MAP sensor

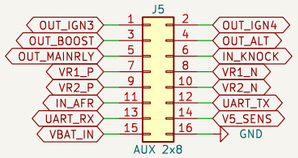


IAT: onboard connector (R4, default) or stock harness (R5)

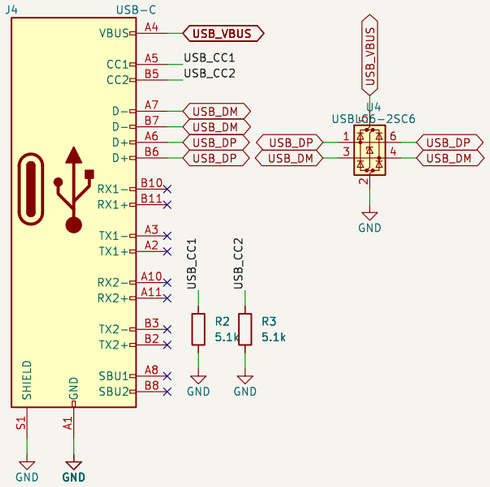


R5 not fitted

AUX: add−on I/O the stock harness has no pins for



USB−C (device): tuning, logs, DFU firmware update



Generated from scripts/mb_nb1.py; netlist identical to the routed board

Sheet: /LSU, MAP, IAT, USB, AUX/		
File: io.kicad_sch		
Title: Miata ECU motherboard 1999–2000 — LSU, MAP, IAT, USB, AUX		
Size: A3	Date: 2026−10−29	**Rev: 0.2**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 4/4